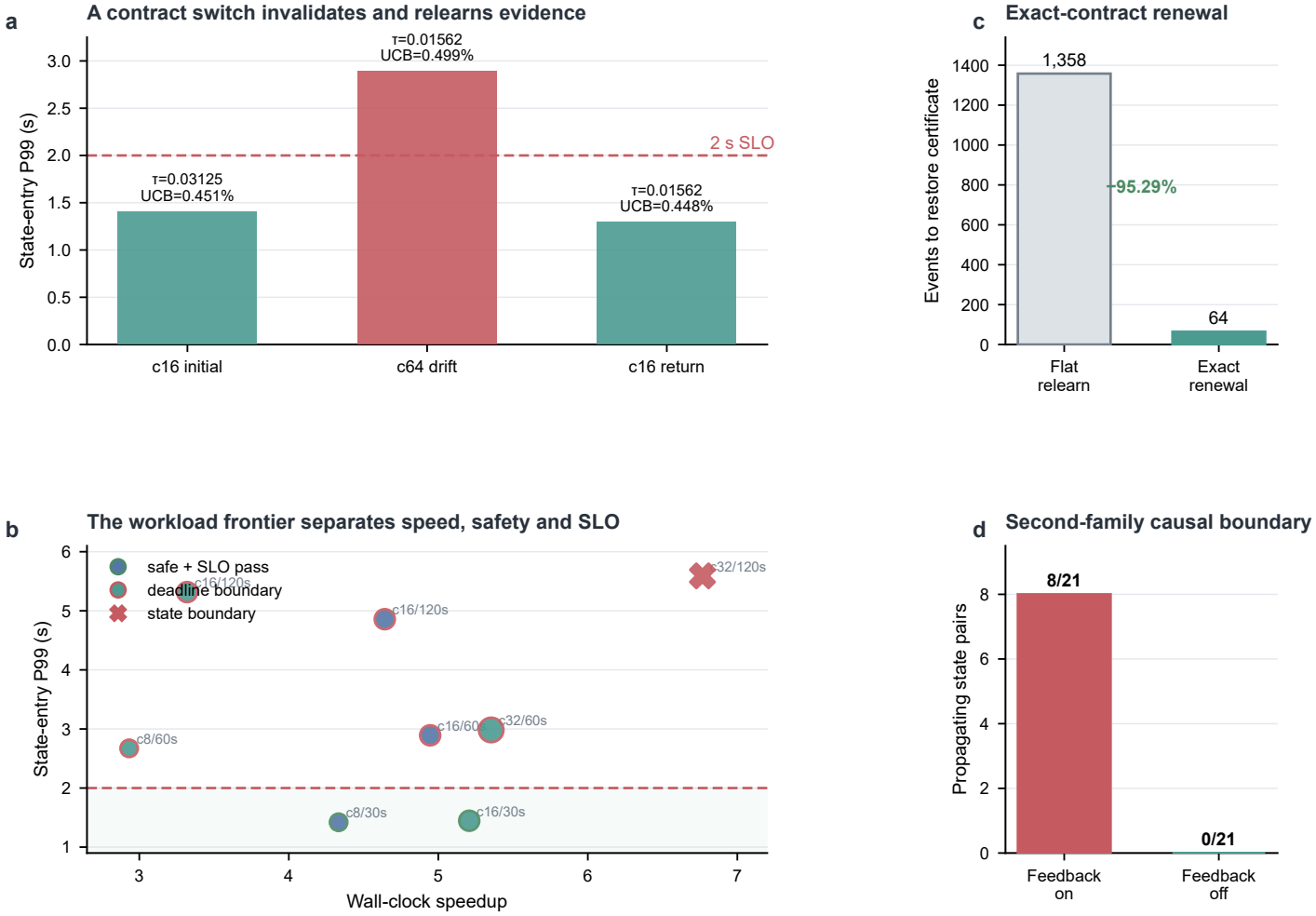

a
A contract switch invalidates and relearns evidence
State-entry P99 (s)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
τ=0.03125
UCB=0.451%
τ=0.01562
UCB=0.499%
2 s SLO
τ=0.01562
UCB=0.448%
c16 initial
c64 drift
c16 return
c
Exact-contract renewal
Events to restore certificate
1400
1200
1000
800
600
400
200
0
1,358
−95.29%
64
Flat
relearn
Exact
renewal
b
The workload frontier separates speed, safety and SLO
State-entry P99 (s)
6
5
4
3
2
1
safe + SLO pass
deadline boundary
state boundary
c16/120s
c16/120s
c32/120s
c8/60s
c16/60s
c32/60s
c8/30s
c16/30s
3
4
5
6
7
Wall-clock speedup
d
Second-family causal boundary
Propagating state pairs
8
6
4
2
0
8/21
0/21
Feedback
on
Feedback
off